

Received 16 November 2024, accepted 16 December 2024, date of publication 20 December 2024,
date of current version 30 December 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3520670

RESEARCH ARTICLE

Dynamic Pricing Strategy for Data Product Through Deep Reinforcement Learning

JUNXIN SHEN¹, YASHI WANG¹, AND FANGHAO XIAO²

¹School of Economics and Management, Kunming University of Science and Technology, Kunming 650500, China

²Marxist College, Xiamen Institute of Technology, Xiamen 361021, China

Corresponding author: Fanghao Xiao (xiaofanghao@xit.edu.cn)

This work was supported in part by the National Natural Science Foundation of China under Grant 72464018, in part by the Ministry of Education of Humanities and Social Science Project under Grant 24YJA630076, in part by the Applied Basic Research Key Project of Yunnan under Grant 202401AS070112, in part by the Science Development Strategy and Policy Research Project of Yunnan under Grant 202404AL030013, and in part by the Provincial-University Collaboration Project of Yunnan under Grant SYSX202409.

ABSTRACT With the rapid development of the data trading market, traditional fixed pricing strategies can no longer effectively reflect the real value of data products, thereby restricting the development of the data trading market. To address this challenge, this paper proposes a dynamic pricing method for data products based on deep reinforcement learning, aiming to attract buyers through dynamic pricing strategies and maximize the cumulative profits of data sellers, thus driving the further development of the data trading market. First, the dynamic pricing problem for data products is modeled as a Markov Decision Process (MDP). Then, a dynamic pricing algorithm based on the Deep Q-learning algorithm is designed, incorporating an annealing mechanism to optimize the exploration strategy. Finally, the performance of traditional reinforcement learning algorithms, specifically Q-learning and SARSA, is compared. The experimental results show that the dynamic pricing method based on deep reinforcement learning not only effectively enhances the cumulative profits of sellers but also significantly improves both revenue and algorithm performance compared to static pricing strategies and traditional reinforcement learning algorithms. This research provides a novel solution for data product pricing, contributing to the healthy development of the digital economy.

INDEX TERMS Data trading market, dynamic pricing, deep reinforcement learning, digital economy.

I. INTRODUCTION

As technology continues to advance, data-driven business models are increasingly emerging, underscoring the growing significance of data products. These products, serving as a core resource, are developed through the processes of data collection, storage, processing, analysis, and visualization to create services or products that meet user needs [1]. As such, data products have become a primary commodity in the data factor market [2]. Data trading platforms, which act as the key intermediaries between data products sellers and buyers while ensuring trust, play a critical role in eliminating data silos and facilitating data circulation [3]. Within this context, the challenge of effective pricing has emerged as a crucial problem that needs to be addressed.

The associate editor coordinating the review of this manuscript and approving it for publication was Wei-Wen Hu¹.

Pricing is fundamental to data element transactions and is critical to the sustainable development of data trading markets [4]. However, due to the unique characteristics of data such as nonrivalry, non-depletability, replicability, time-liness, bidirectional value uncertainty, and near-zero marginal costs [5], [6], the pricing of data products is more complicated than that of other products. Currently, most data pricing methods rely on static pricing, which fails to meet the dynamic adjustment and balancing needs of online transactions, thereby limiting the potential for sellers to maximize long-term cumulative profits [7]. In contrast, dynamic pricing strategies are more likely to enhance the maximization of cumulative profits effectively.

With the continuous advancement of artificial intelligence technology, reinforcement learning has become a crucial tool for addressing dynamic pricing challenges, with Q-learning, SARSA, and their improved algorithms being widely applied.

Reinforcement learning enables the autonomous learning of optimal strategies through ongoing interaction with the environment and trial-and-error, thereby maximizing cumulative returns. Deep reinforcement learning further overcomes the limitations of traditional reinforcement learning, such as slow or impossible convergence. However, research applying deep reinforcement learning to the dynamic pricing of data products remains relatively scarce. In response to this gap, this paper proposes a dynamic pricing model aimed at maximizing the cumulative profits for data product sellers. The problem is modeled as a Markov Decision Process and is solved using deep reinforcement learning methods. The main contributions of this paper are as follows:

- (1) Identification of key characteristics of data products and development of a dynamic pricing model for data products: This paper highlights the unique features of data products in contrast to traditional commodities and proposes a dynamic pricing revenue model that incorporates the time-sensitive nature of data products. The model integrates a discount factor to account for the time-dependent value of data, distinguishing it from traditional goods that typically rely on inventory-based pricing strategies.
- (2) Application of deep Q-learning for dynamic pricing: Leveraging deep reinforcement learning theory, this paper introduces a Deep Q-learning algorithm tailored to address the dynamic pricing challenge for data products. The algorithm optimizes the pricing decisions over time, enabling data sellers to maximize cumulative revenue.
- (3) Innovative solutions for data trading platforms: By applying the Deep Q-learning method to data product pricing, this paper provides innovative solutions for data trading platforms, offering a scalable adaptable pricing strategy that can adjust to market fluctuations. This approach not only broadens the use of dynamic pricing in the context of data products but also contributes to the sustainable growth and expansion of the data trading market.

The rest of this paper is as follows: Chapter 2 reviews the relevant literature. Chapter 3 describes the research problem and models it as a Markov Decision Process. Chapter 4 develops dynamic pricing algorithms for data products based on the Deep Q-learning algorithm in deep reinforcement learning. Chapter 5 presents the experimental design and analyzes the results. Finally, Chapter 6 summarizes the main work of this paper and discusses some limitations.

II. LITERATURE REVIEW

A. CHARACTERISTICS AND STRATEGIES OF DATA PRODUCT

Compared to traditional commodities, data products exhibit a range of unique characteristics, such as non-consumable, replicability, nonrivalry, timeliness, bidirectional value uncertainty, and near-zero marginal cost [5], [6], which make

pricing data more complex than other products [8]. First, non-consumable means that data products do not diminish in quantity or quality during use, and repeated use does not affect their original state. Replicability refers to the ability to infinitely copy and use data products without degrading their quality or integrity. Nonrivalry indicates that one user's consumption does not reduce the availability of the same data product to others, nor does it cause usage conflicts. Furthermore, timeliness highlights that the value of data products changes over time, while bidirectional value uncertainty suggests that the value of data varies depending on the application and user, with different users potentially assigning different values to the same data product. Additionally, near-zero marginal cost means the extra cost of replicating data is almost negligible. Currently, static pricing methods dominate data product pricing, making it difficult to meet the need for dynamic price adjustments and balancing in online transactions. This limitation hinders sellers from maximizing long-term cumulative revenue, thus impeding the sustainable development of data trading [7]. In contrast, dynamic pricing strategies offer the potential to more effectively maximize cumulative revenue, implementing dynamic pricing a critical issue that urgently needs to be addressed.

Currently, pricing strategies for data products can be categorized into five main types: negotiated pricing, auction pricing, usage-based pricing, freemium pricing, and dynamic pricing [9]. Negotiated pricing involves reaching an agreement between buyers and sellers through negotiation, offering high flexibility and enhancing transaction success rates. However, its applicability in large-scale markets is limited due to the time-intensive nature of negotiations and the high resource demands. Auction pricing, which determines prices through a bidding mechanism, is typically suited for scarce or high-value data products. Usage-based pricing charges users based on actual usage or frequency, providing flexible price adjustments but potentially leading to unstable revenue streams. Freemium pricing offers basic data services for free while charging for premium features or additional services, a strategy that can attract a large user base but may face challenges in defining the boundaries between free and paid versions. Dynamic pricing adjusts prices in real-time based on factors such as market demand, supply and demand dynamics, competition, and user behavior, but it requires the support of complex algorithms.

B. DYNAMIC PRICING AND ITS APPLICATION

Dynamic pricing has demonstrated remarkable adaptability and economic benefits across different periods and industries. Its origins and evolution, from traditional market transactions to modern intelligent algorithms, reflect the integration of business strategy and technological advancement. At present, dynamic pricing is widely applied across various fields to adapt to changing market conditions and demands, thereby maximizing revenue and operational efficiency for businesses.

In the context of dynamic pricing for hotel rooms, Vives and Jacob [10] applied online hotel demand functions to deterministic and stochastic dynamic models to determine the optimal room prices that maximize revenue. Bandalouski et al. [11] approached the hotel dynamic pricing issue by decomposing demand and solving it through a dynamic programming model. Tang et al. [12] utilized machine learning techniques to predict Airbnb prices. In the airline industry, Otero and Akhavan-Tabatabaei [13] balanced the trade-off between selling seats at low prices and flying with empty seats by fitting customer purchase probabilities using phase-type distributions, thus establishing a dynamic pricing model for flights. Shihab et al. [14] modeled flight seat inventory and overbooking issues as a Markov decision process and designed a DQN algorithm based on deep reinforcement learning to seek optimal pricing decisions that maximize revenue for each flight. In the ride-hailing sector, Wu et al. [15] considered spatial heterogeneity and network externalities to study optimal pricing for ride-hailing platforms from a two-sided market perspective. Huang et al. [16] tackled the long-term optimization problem of dynamic pricing for ride-hailing platforms using a soft actor-critic algorithm within a deep reinforcement learning framework. In online retail dynamic pricing, Rana and Oliveira [17] constructed dynamic pricing models for multiple interdependent perishable products based on a Markov decision process under stochastic demand with unknown functional forms, using the $Q(\lambda)$ algorithm from reinforcement learning to solve the pricing problem. Dogan and Güner [18] addressed the optimal pricing strategy for retailers in a competitive environment using the Q-learning algorithm from reinforcement learning. Additionally, Kim et al. [19] explored dynamic pricing and energy consumption scheduling for service providers acting as intermediaries between power companies and users within microgrids, using reinforcement learning. Chakole and Kurhekar [20] described a deep Q-learning agent capable of learning trend-following trading strategies and making stock trading decisions based on reward-driven learning. Martín-Perez et al. [21] studied the dynamic pricing of public cloud service providers based on DQN algorithm. Moreover, it also has shown excellent performance in stock price prediction [22], agricultural product price prediction [23], crude oil price prediction [24], and cryptocurrency price prediction [25].

The methodologies employed to address dynamic pricing include mathematical programming and operations research, game theory, reinforcement learning, and deep reinforcement learning. Mathematical programming and operations research methods are typically applied to dynamic pricing problems where resources are constrained and demand environments are known. These methods focus on optimizing seller profits based on known demand functions and resource limitations. However, in the case of data products, the demand environment is often unknown, and the marginal cost approaches zero, making these traditional approaches

less effective for dynamic pricing in this domain. Game theory, on the other hand, is often used in scenarios involving multiple participants, such as bilateral markets or interactions between suppliers and retailers. It analyzes strategic decision-making and outcomes under competitive or cooperative conditions. While game theory is valuable in such contexts, the dynamic and time-varying nature of data product markets makes it difficult to apply to their dynamic pricing challenges. In contrast, deep reinforcement learning, which combines the perceptive capabilities of deep learning with the decision-making capabilities of reinforcement learning, has emerged as one of the most promising areas of artificial intelligence research [26]. It has shown great potential for solving dynamic pricing problems due to its ability to learn from continuous interaction with the environment and adapt to complex, uncertain settings. It represents a likely direction for the future development of automated trading and offers greater scalability, making it highly suitable for dynamic pricing of data products.

C. DEEP REINFORCEMENT LEARNING METHODOLOGIES

The Markov Decision Process (MDP) forms the theoretical foundation for reinforcement learning modeling. It provides a mathematical framework to describe the interaction between the environment and the agent. Modeling sequential decision problems using an MDP is a prerequisite for applying reinforcement learning to solve these problems.

As a subset of machine learning algorithms within the field of artificial intelligence, reinforcement learning (RL) is a goal-directed learning method that trains an agent to maximize cumulative rewards through interaction with its environment. It is primarily applied to solve sequential decision-making problems. The main objective in RL is to find an optimal policy that allows the agent to achieve the maximum cumulative reward over the long term. Two representative algorithms in reinforcement learning are the Q-learning algorithm and the SARSA algorithm.

Q-learning is a model-free, off-policy reinforcement learning algorithm that learns the action-value function, enabling the value function for action selection in each state to gradually approach the optimal value. It is particularly suitable for problems that require exploration during learning and convergence to the optimal policy. The SARSA (State-Action-Reward-State-Action) algorithm, on the other hand, is an on-policy reinforcement learning algorithm that updates the action-value function under the current policy. It is suitable for problems requiring online learning and real-time decision-making.

In the past decade, deep reinforcement learning has been widely applied in areas such as AlphaGo, gaming, and robotics [27], [28], [29], [30]. When dealing with high-dimensional state and action spaces, both the SARSA and Q-learning algorithms are prone to the dimensional disaster [31]. In response, Mnih et al. [32], [33] proposed the Deep Q-Network (DQN) algorithm, which integrates

deep learning with reinforcement learning. The major contributions are twofold: First, it introduced the experience replay, which stores experience samples obtained from the agent-environment interaction in a replay buffer and periodically samples a batch of these experiences with equal probability for Q-network training and parameter updates. It addresses a significant issue in the integration of deep learning and reinforcement learning, specifically the strong correlation in the sequence of experiences generated by the agent-environment interaction, which undermines the assumption of independent and identically distributed data required for deep learning models, thereby affecting performance. Second, DQN employs the fixed target network, which uses a target network with the same architecture as the Q-network to calculate Q-target values and periodically updates the target network by copying parameters from the Q-network, thereby mitigating training instability.

Building on DQN, many scholars have further refined the algorithm. For example, Double DQN [34] combines DQN with Double Q-learning to address the issue of Q-value overestimation in DQN; Dueling DQN [35] improves action-value estimation by decomposing the state-value function into state value and advantage function; Rainbow DQN [36] integrates several improvements of DQN into a single model, among other advancements.

D. LITERATURE REVIEW SUMMARY

In summary, data products possess unique characteristics that differentiate them from other goods, making their pricing highly complex. Theoretically, dynamic pricing strategies are the most suitable approach for addressing the challenges of pricing data products. While deep reinforcement learning has demonstrated superior performance in solving dynamic pricing problems for other goods, research on dynamic pricing strategies for data products remains limited. Therefore, applying deep reinforcement learning to study the dynamic pricing of data products is of great importance. It not only offers a solution to the pricing challenges but also provides new directions for the development of the data trading market.

III. PROBLEM DESCRIPTION AND MODELING

In this section, we will provide a detailed description of the problem and establish the mathematical model of dynamic pricing of data products, then abstractly model it as a Markov Decision Process.

A. PROBLEM DESCRIPTION

In the context of a rapidly changing market environment, setting a fixed price for data products throughout the entire sales cycle does not align with the dynamic nature of data product value and fails to ensure the maximization of cumulative revenue for the data sellers. As illustrated in figure 1, the seller lists the data products on the data trading platform, which then employs a dynamic pricing system to determine and publicly announce an optimal price based on various factors. Potential buyers compare the platform's announced price

with their reservation price. If the announced price exceeds the reservation price, they choose not to purchase; otherwise, they proceed with the purchase, paying the platform for the data product. After deducting its commission, the platform transfers the remaining amount to the seller, completing the transaction.

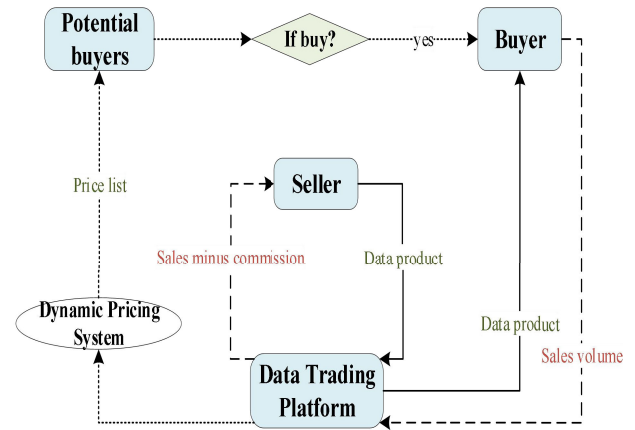


FIGURE 1. Transaction process.

For this transaction process, it is assumed that the total sales period T for the data product is divided into several equal-length sub-periods $t \in [1, 2, 3, \dots, T]$. In each sub-period, the product is priced at $p^{(t)}$, and the sales volume is $d^{(t)}$. The data trading platform charges a commission based on the sales volume at a rate of R_s . Additionally, the seller incurs a fixed cost C_f every sub-period. The seller's revenue objective function can then be expressed as shown below:

$$\max z = \sum_{t=1}^T \gamma^{t-1} \left\{ (1 - R_s) \times p^{(t)} \times d^{(t)} - C_f \right\} \quad (1)$$

B. MODELING

The above transaction process can be modeled as a discrete finite Markov Decision Process (MDP) defined by the five-tuple $M = (S, A, R, P, \gamma)$, where each element is described as follows:

State Space (S): $S = [s^{(1)}, s^{(2)}, \dots, s^{(T)}]$, it represents the set of all possible states of the environment, encompassing all the information the agent can obtain through interactions with the environment. $s^{(t)} \in S$, denotes the environment's state at sub-period t . In this context, $s^{(t)}$ is composed of the price vectors from all previous time steps before the t -th sales sub-period.

Action Space (A): it represents the set of actions the agent can take, which can be either continuous or discrete. In this context, it representing the set of executable pricing actions, the action $a^{(t)}$ denotes the pricing action taken by the agent during the t -th sales sub-period.

State Transition Probability Function (P): it represents the probability of transitioning from the current state to the next state, typically denoted as $P(s' | s, a)$, where s is the current

state, a is the action taken, and s' is the next state. It illustrates the probability of transitioning from state s to state s' after taking action a . Since pricing changes can affect the volume of browsing and buyers' demand response at different prices, this study employs deep reinforcement learning to solve the state transition probability, thereby avoiding the need for a priori knowledge of these probabilities.

Reward Function (R): it represents the feedback the environment provides, typically denoted as $R(s, a)$, which means the reward given by the environment after the agent takes action a in state s . In this context, after the agent takes pricing action $a^{(t)}$ in state $s^{(t)}$ during the t -th sales sub-period, the immediate reward obtained is $r^{(t+1)}$, representing the revenue for that sales period, as shown below:

$$r^{(t+1)} = (1 - R_s) \times p^{(t)} \times d^{(t)} - C_f \quad (2)$$

Discount Factor (γ): it measures the importance of future rewards. A smaller γ value indicates that the agent prioritizes immediate rewards, whereas a larger γ value signifies greater concern for future rewards.

The dynamic pricing process based on the Markov Decision Process is illustrated in Figure 2. During the t sub-period, the agent acquires the current state information and immediately executes the corresponding action. After interacting with the environment, the agent receives an immediate reward and transitions the current state to the next period's state. By continuously iterating this interaction process, the agent utilizes environmental feedback to experiment, accumulate experience, optimize the pricing strategy, and ultimately achieve the goal of maximizing cumulative revenue.

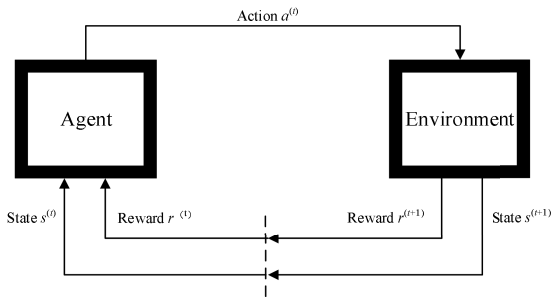


FIGURE 2. The MDP of dynamic pricing.

The mathematical symbols and their meanings are presented in Table 1.

IV. DYNAMIC PRICING ALGORITHM

A. Q-LEARNING

Q-learning is a model-free, off-policy reinforcement learning algorithm that learns the action-value function, enabling the value function for action selection in each state to gradually approach the optimal value. It is particularly suitable for problems that require exploration during learning and convergence to the optimal policy. The Q-value update rule is

TABLE 1. Notation table.

Notation	Parameter Definition
t	the t -th sales sub-period
T	total sales period
$p^{(t)}$	the price in the t -th sales sub-sales
$d^{(t)}$	The sales volume in the t -th sales sub-sales
R_s	commission rate
C_f	fixed costs every sub-sales
γ	discount factor
$s^{(t)}$	the environment state at the t -th sales sub-sales
$a^{(t)}$	The action at the t -th sales sub-sales
$r^{(t)}$	the revenue for sub-period $t-1$

expressed as follows:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left\{ R + \gamma \max_{a'} Q(s', a') - Q(s, a) \right\} \quad (3)$$

where α represents the learning rate, R represents the immediate reward, γ represents the discount factor, s' represents the next state after taking action, $Q(s', a')$ represents the optimal action value in the next state.

The algorithm framework is shown below:

Algorithm 1 Dynamic Pricing Algorithm Based on Q-Learning

```

INPUT: pricing list  $A$ , episode length  $K$ , pricing period  $T$ ,
learning rate  $\alpha$ , discount factor  $\gamma$ 
OUTPUT: optimal pricing strategy from  $t = 0$  to  $t = T - 1$ 
For  $k = 0, K$  do
    Initialize environment
    For  $t = 0, T$  do
        Choose action  $a$  using greedy strategy
        Execute action  $a$ , observe reward  $r$  and next state  $s'$ 
        Update  $Q$ :  $Q(s, a) \leftarrow Q(s, a) + \alpha \left\{ r + \gamma \max_{a' \in A} Q(s', a') - Q(s, a) \right\}$ 
        Update  $s$ 
    end
end

```

B. SARSA

SARSA is an on-policy reinforcement learning algorithm that is suitable for problems requiring online learning and real-time decision-making. The Q-value update rule for SARSA is as follows:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \{ r + \gamma Q(s', a') - Q(s, a) \} \quad (4)$$

where a' represents the next action selected in state based on the current policy.

The algorithm framework is shown below:

C. DEEP Q-LEARNING

Deep Q-learning is a reinforcement learning algorithm that combines deep learning and Q-learning to address Markov

Algorithm 2 Dynamic Pricing Algorithm Based on SARSA

INPUT: pricing list A , episode length K , pricing period T , learning rate α , discount factor γ
 OUTPUT: optimal pricing strategy from $t = 0$ to $t = T - 1$

For $k = 0, K$ **do**
 Initialize environment
 For $t = 0, T$ **do**
 Choose action a using greedy strategy
 Execute action a , observe reward r and next state s'
 Update $Q: Q(s, a) \leftarrow Q(s, a) + \alpha \{r + \gamma Q(s', a) - Q(s, a)\}$
 Update s
 end
end

Decision Processes in continuous state and action spaces. Unlike the Q-learning algorithm, where Q-values are typically represented in tabular form, Deep Q-learning approximates the Q-value table using a neural network and employs mean squared error $L(\theta)$ as the loss function. The Q-network is continuously trained using gradient descent to handle high-dimensional and nonlinear state and action spaces. The formulation of $L(\theta)$ is presented in Equation (5).

$$L(\theta) = E[(r + \gamma \max_{a' \in A} Q(s', a' | \theta) - Q(s, a | \theta))^2] \quad (5)$$

Balancing exploitation and exploration is critical in the iterative training process. Therefore, this paper introduces an annealing mechanism to improve the ε -greedy strategy, making it more suitable for solving the dynamic pricing problem of data products. This strategy begins with a large initial value ε , which decays exponentially as the number of iterations increases. In the early stages, the algorithm emphasizes exploration to avoid getting trapped in local optima, while in the later stages, it focuses more on exploitation of the knowledge gained to accelerate convergence. In each iteration, the algorithm generates a random number η within the range $[0, 1]$. If ε is less than η , the action corresponding to the maximum Q-value is selected; otherwise, a random action is chosen. A smaller value of ε leads to more exploitation, while a larger ε value encourages more exploration.

Based on the principles of the Deep Q-learning algorithm, the algorithm framework for implementing dynamic pricing strategies for data products is proposed, as outlined below.

V. NUMERICAL EXPERIMENTS

In this section, the experimental environment will be set up. Subsequently, experiments will be conducted, and the results will be analyzed.

A. EXPERIMENTAL SETTING

In the experimental environment, the agent autonomously makes pricing decisions based solely on interactions with the environment and historical data. The experimental data is generated based on the following assumptions.

Algorithm 3 Dynamic Pricing Algorithm Based on Deep Q-learning

INPUT: evaluation network Q , target network Q' , pricing list A , replay memory D with capacity N , episode length K , pricing period T , number of steps to update the target network parameter C , learning rate α , discount factor γ .
 OUTPUT: optimal pricing strategy from $t = 0$ to $t = T - 1$

For $k = 0, K$ **do**
 Initialize environment, s
 For $t = 0, T$ **do**
 Choose action a using greedy strategy with an annealing mechanism
 Execute action a , observe reward r and next state s'
 Obtain experience tuple $\langle s, a, r, s' \rangle$ into D
 Sample random experience tuples from D
 Calculate $y^{(t)}$

$$= \begin{cases} r^{(t)}, & \text{if episode terminates at step } t + 1 \\ r^{(t)} + \gamma \max_{a'} Q(s', a' | \hat{\theta}), & \text{otherwise} \end{cases}$$

 Perform gradient descent
 Update s
 Reset $Q' = Q$ every C steps
 end
end

The traditional supply-demand relationship is typically used for pricing conventional products, where supply constraints play a key role in determining optimal pricing. However, data products do not have the same inventory limitations, making it infeasible to apply a traditional supply-demand model for pricing. In this paper, we propose a new pricing way that focuses on how price fluctuations influence consumer purchasing behavior, which in turn affects the sales volume of data products. The following is the specific experimental setup.

First, regarding customer browsing volume, two factors are considered. The first factor is that, in the absence of price fluctuations, customer demand for data products decreases over time and follows a non-homogeneous Poisson distribution with intensity $\lambda(t) = \lambda_0 - d \times t$. The second factor considers the impact of price fluctuations, where the effect of a price increase is represented by the function $h_1(p^{(t)}, p^{(t-1)}) = \chi_1 \sqrt{p^{(t)} - p^{(t-1)}}$ with an impact coefficient χ_1 , and the effect of a price decrease is represented by the function $h_2(p^{(t)}, p^{(t-1)}) = \chi_2 \sqrt{p^{(t-1)} - p^{(t)}}$ with an impact coefficient χ_2 . Therefore, the customer browsing volume $B^{(t)}$ in period t can be expressed as shown below:

$$B^{(t)} = \begin{cases} \lambda^{(t)} - \chi_1 \sqrt{p^{(t)} - p^{(t-1)}}, & p^{(t)} \geq p^{(t-1)} \\ \lambda^{(t)} + \chi_2 \sqrt{p^{(t-1)} - p^{(t)}}, & p^{(t)} \leq p^{(t-1)} \end{cases} \quad (6)$$

Note: when $B^{(t)} < 0$, set $B^{(t)} = 0$.

Second, regarding the probability of a customer making a purchase, it is assumed that the customer's reservation price follows an exponential distribution with parameter v . When the product price is $p^{(t)}$, the probability

TABLE 2. Specific setting of experimental parameters.

Notation	Parameter Definition
T	20
A	$\{20,21,22,\dots,98,99\}$
R_s	10%
C_f	1000
λ_0	1000
h	50
x_1	30
x_2	50
v	50
N	100000

distribution function of a customer choosing not to purchase is $F(p^{(t)}) = 1 - e^{-p^{(t)}/v}$.

In summary, the sales volume $d^{(t)}$ of data products in period t can be expressed as the product of the customer browsing volume and the probability of the customer choosing to purchase, as shown below:

$$d^{(t)} = B^{(t)} \times (1 - F(p^{(t)})) \quad (7)$$

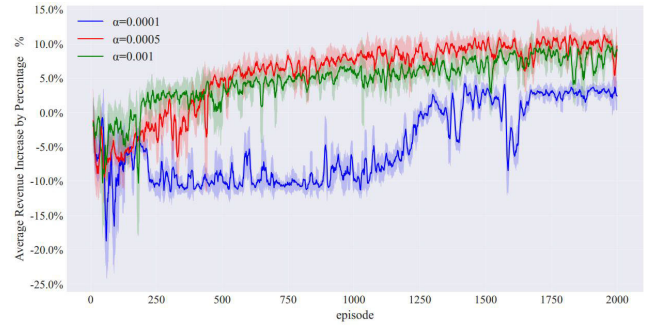
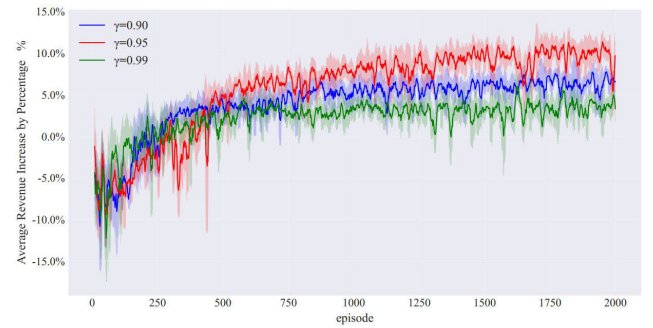
The general parameter settings for the experiment are listed in Table 2.

B. EXPERIMENTAL RESULTS

After setting up the experiment, experiments were conducted, and the performance of the algorithm was analyzed based on the numerical results obtained through interaction. The experimental content includes two parts: the first part involves tuning the hyperparameters of the proposed algorithm using the control variable method. Specifically, we analyze the impact of learning rate α , discount factor γ , and exploration decay rate r_d on the experimental results, with the goal of identifying the hyperparameter combination that yields the most significant improvement over the optimal fixed pricing. This improvement is measured by the “Average Revenue Increase by Percentage,” which indicates the percentage increase in achieved through dynamic pricing as compared to a fixed pricing strategy, and is shown in Figures 3-5 and Figure 6 (b). The second part compares the performance of the algorithm with Q-learning and SARSA algorithms.

When selecting hyperparameters, it is essential to consider both the stability and convergence speed of the algorithm. Therefore, experiments were conducted to evaluate the effects of different hyperparameter combinations for learning rate α , discount factor γ , and exploration decay rate r_d , aiming to find the optimal configuration for the algorithm.

Learning Rate α : the learning rate α controls the step size of weight updates during each training iteration. Selecting an appropriate learning rate is crucial for the algorithm’s performance and convergence speed. A higher learning rate can accelerate the algorithm’s convergence but may cause the loss function to overshoot the global optimum, missing the best solution; a lower learning rate, while conducive to finding the

**FIGURE 3.** Percentage Improvement in Average Revenue under Different α .**FIGURE 4.** Percentage Improvement in Average Revenue under Different γ .

global optimum, results in slower convergence and requires more training time. In this study, with γ fixed at 0.95 and r_d at 0.0025, three comparative experiments were conducted with α set to 0.0001, 0.0005, and 0.001, respectively. The experimental results, as shown in Figure 3, indicate that the algorithm achieves the most significant improvement in average profit percentage when α is set to 0.0005.

Discount Factor γ : The discount factor γ influences the degree to which the algorithm considers the Q-value of the next state when updating the current Q-value. An appropriate γ value can balance the trade-off between immediate and future rewards. When γ is set too high, the algorithm emphasizes future rewards; conversely, when γ is set too low, the algorithm prioritizes immediate rewards. In this study, with α fixed at 0.0005 and r_d at 0.0025, three comparative experiments were conducted with γ set to 0.9, 0.95, and 0.99, respectively. The results, as shown in Figure 4, indicate that the algorithm achieves the most significant improvement in average profit percentage when γ is set to 0.95.

Exploration Decay Rate r_d : The exploration decay rate is used to adjust the exploration rate ε in the ε -greedy strategy, determining how quickly ε decreases as the number of iterations increases. A larger r_d will cause ε to decrease more rapidly, leading the algorithm to favor exploitation of the knowledge already acquired; conversely, a smaller r_d will slow down the decrease in exploration rate, thereby emphasizing exploration of the unknown environment. Choosing an appropriate r_d allows for a better balance between exploration

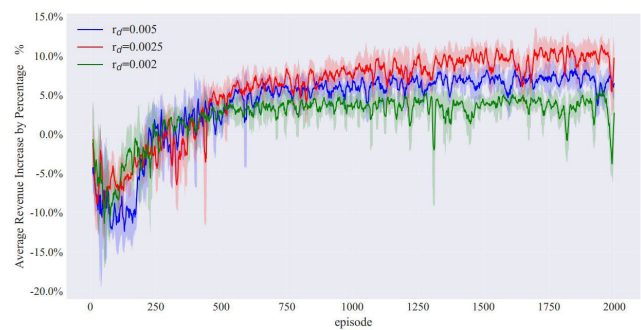


FIGURE 5. Percentage Improvement in Average Revenue under Different r_d .

TABLE 3. The variation in average profit with the number of iterations under the optimal parameters.

$k(\text{episode})$	Percentage(%)	$k(\text{episode})$	Percentage(%)
100	82.80%	200	83.24%
300	84.40%	400	84.96%
500	86.07%	600	87.17%
700	88.09%	800	88.79%
900	89.37%	1000	89.90%
1100	90.39%	1200	90.79%
1300	91.13%	1400	91.48%
1500	91.81%	1600	92.09%
1700	92.29%	1800	92.56%
1900	92.77%	2000	92.95%

and exploitation, resulting in improved algorithm performance. In this study, with α fixed at 0.0005 and γ at 0.95, three comparative experiments were conducted with r_d set to 0.005, 0.0025, and 0.002, respectively. The results, as shown in Figure 5, indicate that the algorithm achieves the most significant improvement in average profit percentage when r_d is set to 0.0025.

The experimental results demonstrate that the parameter combination of $\alpha = 0.0005$, $\gamma = 0.95$, and $r_d = 0.0025$ achieves the best performance in terms of profit improvement. As shown in Table 3, when the optimal parameter combination is applied, the percentage of average profit relative to the optimal pricing profit gradually increases with the number of interactive iterations with the environment. As the iterations continue to increase, the average profit keeps rising, eventually leading to the convergence of the strategy and stabilization of the profit.

The next step involves a comparative analysis between the Deep Q-learning-based algorithm presented in this study and the Q-learning and SARSA algorithms. Both Q-learning and SARSA are reinforcement learning methods, with hyperparameters set identical to those optimal parameter combination. The experimental results of different algorithms are shown in Figure 6 and detailed in Table 4. The results indicate that the algorithm based on Deep Q-learning achieves the best performance in terms of revenue, followed by the Q-learning algorithm. In terms of stability, the Q-learning and SARSA algorithms exhibit more fluctuations.

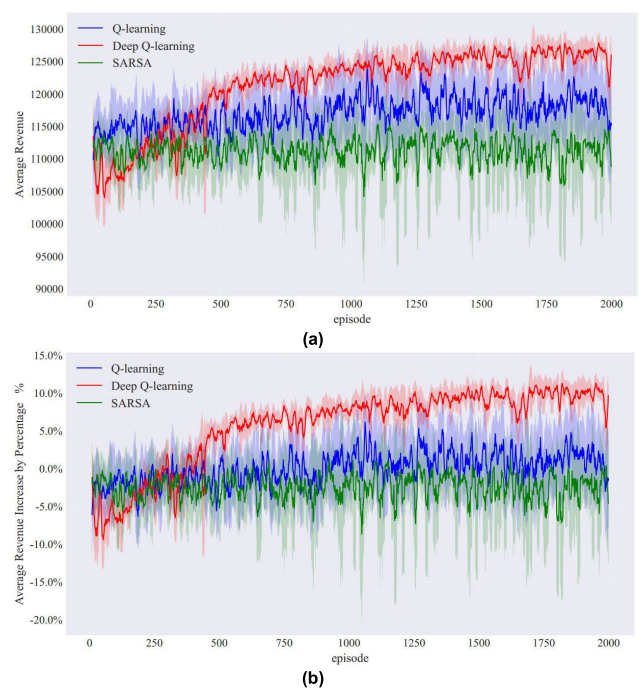


FIGURE 6. Comparison of experimental results of different algorithms.

TABLE 4. Comparison of revenue performance across different algorithms.

Algorithm	Deep Q-learning	Q-learning	SARSA
Min	97483.19	99376.64	79648.25
First quartile	119084.36	112869.87	109201.37
Median	123358.94	117036.87	112127.00
Mean	121348.18	117030.94	111249.89
Third quartile	125805.64	121402.45	114731.79
Max	130547.90	129392.58	124423.73

TABLE 5. Optimal dynamic pricing strategy.

t	$p^{(t)}$	t	$p^{(t)}$
1	86	2	64
3	53	4	44
5	39	6	71
7	44	8	39
9	98	10	53
11	43	12	29
13	98	14	53
15	47	16	26
17	98	18	53
19	35	20	21

Finally, the dynamic pricing strategy over the period is summarized and listed in Table 5.

VI. CONCLUSION

Currently, many data trading platforms face a series of practical issues in achieving further development, with data product pricing being particularly prominent. This paper explores the application of Deep Q-learning to the dynamic pricing problem of data products. Through the combination of

reinforcement learning and deep learning, Deep Q-learning effectively addresses the challenges of real-time pricing in data marketplaces.

Considering the characteristics of data products, such as their timeliness, bidirectional value uncertainty, and near-zero marginal cost, this study first models the problem by constructing a mathematical model for the dynamic pricing of data products: customer browsing volume is simulated using a non-homogeneous Poisson process, and customer purchase decisions are judged based on their reservation price. The platform service fee is calculated based on the sales volume, from which the revenue and cost of each period are derived. These periodic profits are accumulated using a discount rate, forming an overall revenue model. Next, the mathematical problem is converted into a Markov decision process, consisting of five elements: state space, action space, reward function, state transition probability function, and discount factor, enabling it to be solved via reinforcement learning. What is more, a dynamic pricing algorithm for data products is designed using the Deep Q-learning algorithm from deep reinforcement learning to solve the problem. Finally, a comparative analysis is conducted between the Deep Q-learning algorithm and the traditional Q-learning and SARSA algorithms.

The results indicate that the Deep Q-learning algorithm based on deep reinforcement learning effectively addresses the dynamic pricing problem for data trading platforms and significantly enhances the cumulative revenue compared to static pricing strategies. Compared to Q-learning and SARSA algorithms, deep reinforcement learning demonstrates superior pricing strategies and higher effectiveness in the dynamic pricing problem.

The dynamic pricing model and algorithm proposed in this paper enable dynamic pricing for data products, increase the seller's cumulative revenue, and promote the sustainable development of the data trading market. However, there are several limitations in this study. First, the model does not account for the impact of competition or cooperation between data trading platforms on pricing strategies. In addition to the game-theoretic relationship between platform pricing and consumer price expectations, the competitive or cooperative dynamics between platforms could also play a significant role in pricing decisions. Second, the study does not consider scenarios involving buyer returns, which may influence potential buyers' price expectations and, in turn, affect the dynamic pricing strategies of data trading platforms. In future research, we will explore how to incorporate additional market factors while considering the competition and cooperation between platforms. This will allow us to further refine dynamic pricing strategies for data products and extend the applicability of the model to various real-world scenarios.

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ests include the digital economy and decision analysis.

JUNXIN SHEN was born in Fujian, China, in 1978. He received the bachelor's degree from Jiangxi University of Finance and Economics, in 2000, the master's degree from Yunnan University, in 2005, and the Ph.D. degree from Kunming University of Science and Technology, in 2012.

Since 2012, he has been working with Kunming University of Science and Technology, where he is currently working as a Professor with the School of Management and Economics. His research interests include the digital economy and decision analysis.



YASHI WANG was born in Sichuan, China, in 1999. She received the bachelor's degree from Chongqing University of Technology, in 2021. She is currently pursuing the master's degree in management with Kunming University of Science and Technology. Her research interests include digital economy and dynamic pricing in data products.



FANGHAO XIAO was born in Fujian, China, in 1988. He received the bachelor's degree from Huaqiao University, in 2006. He is currently pursuing the master's degree with Kunming University of Science and Technology.

Since 2010, he has been employed with Xiamen Institute of Technology, where he is currently working as an Associate Researcher with the School of Marxism. His research interests include enterprise business management, college education and management, and mental health education.

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